

Sign and Conjugate Errors

Finding and fixing sign and conjugate errors

Directions

- Four of these problems show a student's work that contains exactly one mistake. For those, do both jobs: work the problem correctly yourself, and then say what the student did wrong.
- A correction with no diagnosis earns half the marks. Name the step and the rule, not just the right answer.
- Write every value in standard form $a + bi$, and remember that $i^2 = -1$.

1. Simplify: $(6 - 5i)(6 + 5i)$

Answer: _____

2. Simplify: $(9 + 4i) - (3 - 7i)$

Answer: _____

3. A student wrote:

$$(8 + 2i) - (5 - 6i) = 8 - 5 + 2i - 6i = 3 - 4i$$

(a) Work the subtraction correctly.

Answer: _____

(b) Name the error in the student's line.

Answer: _____

4. Simplify: $\frac{10}{1 + 3i}$

Answer: _____

5. A student's work for $\frac{14 + 8i}{3 - 2i}$:

$$\frac{14 + 8i}{3 - 2i} \cdot \frac{3 + 2i}{3 + 2i} = \frac{26 + 52i}{9 - 4} = \frac{26 + 52i}{5}$$

- (a) Work the division correctly.

Answer: _____

- (b) Name the error. The numerator is right, so say which value is wrong and what it should be.

Answer: _____

6. A student expanded a square like this:

$$(5 - 2i)^2 = 5^2 + (2i)^2 = 25 - 4 = 21$$

- (a) Expand the square correctly.

Answer: _____

- (b) Name the error.

Answer: _____

7. Simplify: $(3 - 4i) - (-6 + i) + (2 - 5i)$

Answer: _____

8. Here is a student's complete solution to $\frac{20 + 15i}{2 + i}$.

Line 1: multiply above and below by $\frac{2 - i}{2 - i}$

Line 2: $= \frac{20 + 15i}{(2 + i)(2 - i)} = \frac{20 + 15i}{5}$

Line 3: $= 4 + 3i$

(a) Give the corrected quotient.

Answer: _____

(b) Say which line is the first one that is wrong, and state the rule it breaks.

Answer: _____